

Zixun Huang

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RESEARCH INTERESTS

Theoretical foundations of machine learning, with an emphasis on **optimization theory and scaling laws**, **LLM reasoning and post-training**, and **applications of AI**.

EDUCATION

University of Pennsylvania, Wharton School <i>Ph.D. in Statistics and Data Science</i>	Aug. 2026 – Present
Peking University, School of Mathematical Sciences <i>B.S. in Statistics; Elite Undergraduate Program for Applied Mathematics; GPA: 3.7/4.0</i>	Sept. 2022 – Jun. 2026
University of California, Berkeley <i>Visiting Student; GPA: 4.0/4.0</i>	Jan. 2025 – Aug. 2025

WORKING EXPERIENCE

Tencent Hy – Qingyun Talent Program <i>Research Intern, Beijing</i>	Jun. 2026 – Aug. 2026
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PUBLICATIONS

- [1] Zhongzhi Li, Yucheng Shi, Zongxia Li, Ruhan Wang, Anhao Li, **Zixun Huang**, Junyao Yang, Lei Ke, Ninghao Liu, Haitao Mi, Leowei Liang. *Recursive Synthesis for Long-Horizon Terminal Tasks*. **Preprint**. [\[Paper\]](#) [\[Hugging Face\]](#) [\[Website\]](#)
- [2] **Zixun Huang***, Jiayi Sheng*, Zeyu Zheng. *Variance-Aware Baselines and Adaptive Learning Rates for Reinforcement Learning with Verifiable Rewards*. **Under review**. [\[Paper\]](#)
- [3] Yuhao Sun, Zekun Wu, **Zixun Huang**, Peijie Zhou. *TracingFlow: A Simulation-Free Trajectory Inference Framework Based on Second-Order Dynamics*. **Under review**.
- [4] Binghui Li*, Fengling Chen*, **Zixun Huang***, Lean Wang*, Lei Wu. *Functional Scaling Laws in Kernel Regression: Loss Dynamics and Learning Rate Schedules*. **NeurIPS 2025 Spotlight**. [\[Paper\]](#) * Equal contribution.

RESEARCH EXPERIENCE

University of California, Berkeley <i>Research Intern, Berkeley, CA</i> <i>Supervisor: Associate Professor Zeyu Zheng</i>	Jan. 2025 – Aug. 2025
Peking University <i>Research Intern, Beijing</i> <i>Supervisor: Assistant Professor Lei Wu</i>	May 2023 – Aug. 2026

SELECTED HONORS AND AWARDS

Grand Prize, “Challenge Cup” Competition	2026
Weiming Scholar, Peking University	2026
Hong Sheng Scholarship	2024
First Prize, Chinese Mathematics Contest	2024
Yau Contest Groups Prize	2024
Xiaomi Scholarship	2023
Gold Prize, Chinese Mathematical Olympiad	2021
Silver Prize, Chinese Mathematical Olympiad	2020

SKILLS

Programming:	Python (NumPy, pandas, scikit-learn, PyTorch), C++, MATLAB, L ^A T _E X
Language:	GRE V160/Q170; GRE Math 970 (95th percentile); TOEFL 101